Intergenerational Housing Misallocation and Fertility

Preliminary Note

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Abstract

Children increase a household's demand for living space. Yet large rental units are scarce, buying a family-sized home requires liquid wealth and access to credit, and older households own much of the stock of large homes. I study whether this allocation of housing raises the cost of having children for young households. I first develop a simple overlapping-generations model in which rental limits, down-payment constraints, and the retention decisions of older owners determine who occupies family-sized housing. I then study the same forces in a finite-life model with income risk, saving, tenure choice, transaction costs, and bequests. The model distinguishes changes in family size among current residents from changes in local births due to entry into the housing market. The preliminary calibration suggests that restricted access to family-sized housing matters for households deciding whether and when to have children, and that home purchases and births are closely connected near the end of the fertile years.

Status of this draft

This is a preliminary note. The analytical results below state the assumptions needed for each derivation. The quantitative results use the July 9, 2026 calibration and should not yet be interpreted as final estimates. I have since corrected the timing with which current housing choices enter market demand and the moments used in estimation. Because the correction changes several target moments, the model must be re-estimated before the policy magnitudes can be evaluated.

The specification below matches the one used for the July estimates: the owner premium applies to usable housing space, bequest utility varies with family size, and all owners face the same debt limit. The July estimates also use the income profile and wealth grid described below. These settings have now been restored in the main code; they do not change the estimates reported here but allow the July exercise to be reproduced. The policy results will be updated after re-estimation.

1 Introduction

A child changes a household's housing problem. It is not only another set of goods to buy, or more time to spend at home. It is often a need for a different dwelling: another bedroom, a larger living space, or a unit where raising a child is feasible. In many high-cost cities this margin is hard to adjust. Large rental units are scarce, family-sized homes are concentrated in ownership, and ownership requires down payments and borrowing capacity at ages when many households have little wealth.

This paper studies whether those space and financial constraints affect final family size and the number of births generated by a cohort. That distinction matters. A housing shock can move births across years without changing completed fertility. It can also change local births by changing who enters a high-cost housing market, even if completed fertility among incumbent households moves little. The question here is when housing constraints change completed fertility, when they mainly change entry and local scale, and how both effects depend on who can access family-sized space over the lifecycle.

The mechanism has three parts. First, children raise the household's demand for space. If the rental market does not offer sufficiently large units, the shadow price of an additional room can exceed the posted rent. Second, the natural substitute is ownership, but buying a family-sized home requires liquidity and credit. A young household may value the larger unit and still be unable to purchase it. Third, older owners may remain in large homes even when their own marginal value of space is low, because transaction costs and tax rules subsidize staying put. Housing scarcity then shows up not only as a high price of space, but also as a mismatch between who occupies family-sized units and who values them for fertility.

I begin with a compact two-period model. It shows how rental limits, down-payment constraints, and old-owner retention raise the value of family-sized space for constrained young households and, through that channel, raise the cost of having children. I then embed the same economics in a finite-life quantitative model. The model combines lifecycle fertility choice (Sommer, 2016; Doepke and Kindermann, 2019) with housing tenure choice, collateral constraints, and owner-occupied housing (Henderson and Ioannides, 1983; Sommer et al., 2013; Sommer and Sullivan, 2018). The analytical housing model follows the one-market floorspace logic in Coven et al. (2025): all housing services clear in one market, while tenure, borrowing, and tax rules determine who can actually occupy family-sized units.

I use the current calibration for two purposes. First, I ask whether the model places a high value on family-sized space for households deciding whether to have children. Second, I conduct a preliminary study of housing policies that could move large homes toward younger, constrained households. The calibration finds many renters at the size limit when they are choosing family size and produces a close connection between births and purchases near the end of the fertile window. It also understates early ownership, overstates young liquid wealth, and produces too little exit from ownership in old age. The current exercises therefore speak to the economic mechanism rather than to its final magnitude.

2 Related Literature

The paper connects work on fertility choice to work on housing tenure and the allocation of large homes. Since Becker (1960) and Becker and Lewis (1973), economics has treated children as choice variables that enter preferences and require resources. Quantitative versions place that choice in lifecycle environments with income risk, saving, timing, and intrahousehold bargaining (Jones et al., 2010; Sommer, 2016; Doepke and Kindermann, 2019). Recent work emphasizes that the U.S. fertility decline is difficult to explain by postponement alone or by a temporary shock (Kearney et al., 2022).

Housing prices, housing wealth, and mortgage finance also move fertility behavior. Dettling and Kearney (2014) find opposite house-price effects for owners and nonowners, Lovenheim and Mumford (2013) find that housing wealth raises homeowners' fertility, and Cumming and Dettling (2024) show that mortgage-rate pass-through affects births through cash flow. Hacamo (2021) and Dettling and Kearney (2025) connect mortgage access to home purchases and childbearing. In equilibrium, Couillard (2025) show that the supply of large units matters for aggregate fertility.

The outcome here is closer to completed fertility than to a current birth rate. Simon and Tamura (2009) find that rents per room predict lower fertility, including in completed-fertility specifications. Clark (2012) provides an important caution: expensive housing markets delay first births but do not clearly reduce completed fertility. More recent evidence links housing access to both timing and final family size (Japaridze and Sayour, 2024; Dettling and Kearney, 2025; van Doornik et al., 2025). I use this evidence to distinguish current births, completed fertility among incumbent households, and cohort births after housing-market entry responds.[1]

The closest housing-allocation paper is Coven et al. (2025). Their model studies how property taxes, capitalization, and financial constraints allocate housing between older homeowners and younger constrained households. I add children to that allocation problem. Family-sized space held by an incumbent can then change the private price of children for a young household. The retention force also relates to mortgage lock-in and housing ladders (Fonseca and Liu, 2023; Fonseca et al., 2026). My baseline quantitative model uses transaction costs rather than fixed-rate mortgage lock-in, but the allocation logic is similar.

The quantitative housing model builds on work in which tenure, collateral constraints, and owner-occupied housing shape household behavior (Henderson and Ioannides, 1983; Sommer et al., 2013; Sommer and Sullivan, 2018). It also relates to spatial work in which housing supply and local prices allocate households across markets (Rosen, 1979; Roback, 1982; Gyourko et al., 2013; Hsieh and Moretti, 2019). The present model keeps one housing market so that tenure, family size, and the allocation of space remain the central margins.

3 Analytical Model

3.1 Environment

The economy is populated by overlapping generations of young and old households. A young household has type $i = (y_i, b_i)$: income and wealth at entry. Wealth is inherited from the previous old generation[2]. A household decides whether to enter the economy; inside, she chooses a tenure (rent or own), a house size, completed fertility, and savings for old age. Old households, indexed by j, are of two kinds. Old incumbents, last period's young owners, who enter old age with financial wealth, the house they bought, and a possible accrued gain, choose how much of the house to keep and how large an estate to leave, then exit. The other type is old renters, last period's young renter entrants, who rent in old age and carry no wedge. Nonentrants stay outside the market in both periods. Each period a mass $\bar{M}$ of potential young households enters the pool: the maturing children of the previous cohort plus a renewal inflow. Types are distributed according to an exogenous G_Y.

Households have preferences over consumption, housing, and children. Children raise the household's consumption and housing needs: each child claims $\chi > 0$ units of consumption and $\kappa > 0$ units of

[1]Life-course evidence documents that fertility, homeownership, dwelling size, and residential transitions move together (Mulder and Billari, 2010; Kulu and Steele, 2013; Chudnovskaya, 2019). The model treats those joint transitions as economic choices over the lifecycle.

[2]Wealth is distributed independently of the identity of the household leaving the bequest, although its distribution may be unequal.

space, so with consumption spending c_i, housing h_i, and fertility n_i,

$$u_i^Y = \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i.$$

Old households value consumption, housing, and the bequest $e_j \geq 0$ they leave at exit:

$$u^2 = \log c_j^2 + \gamma \log h_j^2 + \mathcal{B}(e_j),$$

with $\mathcal{B}$ a warm-glow function, increasing and concave, possibly zero, common to all old households. The exiting cohort's estates are what the arriving cohort inherits.

There is a fixed stock $\bar{H}$ of divisible floorspace, held by old incumbents and by passive absentee landlords: buyers purchase from either, renters occupy landlord space, and rents and sale proceeds are transfers.[3] Renting and owning are occupancy modes with different size limits, the largest sizes available to rent and to own,

$$h \leq \bar{h}^R \text{ (rent)}, \qquad h \leq \bar{h}^O \text{ (own)}, \qquad \bar{h}^R < \bar{h}^O,$$

Family-sized houses are mostly available to buy, not to rent. All floorspace trades in one market: the asset price is P, and no-arbitrage ties the rental rate to it,

$$q = \rho P, \qquad \rho = r + \delta + \tau^p.$$

The renter pays q as rent; the owner pays it as mortgage interest, property tax, depreciation, and the forgone return on her equity—different payments, equal per-unit value by arbitrage. Given q, a higher property tax lowers P and with it the down payment on any given house.

A buyer finances at most a share $\phi \in (0,1)$ of the purchase, so she posts a down payment $(1-\phi)Ph$ at purchase, out of her liquid wealth. Households can save in an exogenously provided bond, but renters cannot borrow. The interest rate r is exogenous.

Two taxes appear: the property tax τ^p, part of the user cost above, and the capital-gains tax. Selling a house triggers a tax: the seller pays the rate τ^g on the difference between the sale price and what she paid, in dollars. Dying instead passes the house with a stepped-up basis—the heir's purchase price is reset to the current value—so the gain is never taxed.

Where do gains come from when the real price never moves? From inflation. Dollar prices and incomes all grow at rate ϖ, so relative prices and real choices are unaffected; but the tax form compares dollars from different dates, and a house bought one period ago shows a paper gain even though its real value is unchanged. The tax on that paper gain is real,

$$\bar{\ell} = \tau^g \, \frac{P \, \varpi}{1 + \varpi}$$

per unit sold, and this is the model's only contact with inflation. Every cohort carries the same gain, so every incumbent's unit bears $\bar{\ell}$ if sold while alive.[4] An estate tax, by contrast, would fall on the transfer in any form and leave the sell-or-keep margin undistorted; the gains tax distorts it precisely because death forgives it. Everything is deterministic: a buyer knows the tax her future self will face.[5]

[3]The quantitative model below allows upward-sloping supply; the fixed stock is used here to isolate the allocation of existing space.

[4]The statutory exclusion, which exempts gains below a dollar threshold, is an institutional detail I suppress; with it, small holdings escape the tax and lock-in sorts by size.

[5]Inflation generates nominal taxable gains in an otherwise stationary real environment. Coven et al. (2025) instead generate gains through real growth.

3.2 Households

Conditional on entry, household i chooses a tenure mode $m \in \{R, O\}$, housing, fertility, consumption and savings; old age is discounted at β. For $m \in \{R, O\}$,

$$W_i^m = \max_{c_i, h_i, n_i, b_i'} \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i \\ + \beta\big[\mathbf{1}\{m = R\}\, V^R(b_{j'}) + \mathbf{1}\{m = O\}\, V^O(b_{j'}, h_i)\big]$$

subject to

$$c_i + q h_i + b_i' = y_i + b_i, \qquad h_i \leq \bar{h}^m, \qquad b_i' \geq 0,$$
$$\mathbf{1}\{m = O\}\,(1 - \phi) P h_i \leq b_i.$$

Financial wealth carried into old age is $b_{j'} = (1 + r)\, b_i' - \mathbf{1}\{m = O\}\, q h_i$. Children have a market price: each child claims χ goods and κ units of space, and space rents at q, so providing for a child costs $\chi + \kappa q$ per period. A useful benchmark is equal scaling, $\chi = \kappa q / \alpha$: the child splits her cost between space and goods in the proportion $\kappa q / \chi = \alpha$, the same proportion in which her parents split their own spending.

The old incumbent enters with financial wealth b_j, housing H_j^0, and an accrued gain; old households have no income. She faces a choice with respect to her housing: sell now and pay the tax, or keep it until death, when the tax is forgiven.[6] She solves

$$V^O(b_j, H_j^0) = \max_{c_j^2, h_j^2, e_j} u^2(c_j^2, h_j^2, e_j) \\ \text{subject to} \quad c_j^2 + e_j + q\, h_j^2 = b_j + q H_j^0 - \bar{\ell}\,(H_j^0 - h_j^2), \qquad 0 \leq h_j^2 \leq H_j^0.$$

Selling a unit nets $q - \bar{\ell}$; hence the stepped-up basis acts as a subsidy to locking in. An old renter has no carried housing and no tax:

$$V^R(b_j) = \max_{c_j^2, h_j^2, e_j} u^2(c_j^2, h_j^2, e_j) \quad \text{subject to} \quad c_j^2 + e_j + q\, h_j^2 = b_j, \qquad 0 < h_j^2 \leq \bar{h}^R,$$

and at an interior choice her marginal value of space is exactly q.

Using $P = q/\rho$, owner housing is bounded by

$$H_i^O = \min\left\{\bar{h}^O,\ \frac{b_i\, \rho}{(1 - \phi)\, q}\right\}, \qquad H_i^R = \bar{h}^R.$$

H_i^m is the effective family-housing cap: a renter is capped by the stock, a buyer by the size limit or by collateral, whichever is shorter. Tenure is $W_i^Y = \max\{W_i^R, W_i^O\}$. Entry compares the inside value with an outside option $\bar{W}^E$ under extreme-value shocks of scale κ_E:

$$\pi_i^E = \frac{\exp\{W_i^Y / \kappa_E\}}{\exp\{W_i^Y / \kappa_E\} + \exp\{\bar{W}^E / \kappa_E\}}$$

is the probability that household i enters.

[6] The quantitative model below makes owner housing discrete, although its benchmark omits capital-gains taxation.

3.3 Equilibrium

Aggregate demand adds young households, old incumbents, and old renters, and the single market clears:

$$\bar{M}\int \pi_i^E\, h_i\,\mathrm{d}G_Y(i) + \int h_j^2\,\mathrm{d}F_O(j) + \int h_j^2\,\mathrm{d}F_R(j) = \bar{H},$$

where F_O and F_R are the measures of old incumbents and old renters. A stationary equilibrium consists of a user cost q (and $P = q/\rho$), young policies and entry probabilities, old policies, and stationary measures $(\bar{M}, G_Y, F_O, F_R)$ such that:

1. the policies solve the household problems above at q;
2. the floorspace market clears, as displayed;
3. the population reproduces itself: young owners become the old incumbents, young renters the old renters, and births plus the renewal inflow recreate the cohort;

3.4 Equilibrium policies

The Lagrangian for a young agent is

$$\begin{aligned}\mathcal{L} = u_i^Y + \beta\, V_{j'} + \lambda_i\big(y_i + b_i - c_i - qh_i - b_i'\big)\\ + \mu_i\big(b_i - (1-\phi)Ph_i\big) + \theta_i\big(\bar{h}^m - h_i\big) + \xi_i\, b_i',\end{aligned}$$

with $\mu_i = 0$ in the rental mode and $\xi_i \geq 0$ the multiplier on the no-borrowing bound $b_i' \geq 0$ for a renter. The formulas below apply at the stated interior saving and retention margins; at a corner, the corresponding complementary-slackness condition replaces the equality.

The first-order condition for saving sets the marginal value of resources today against the return tomorrow:

$$\frac{1}{c_i - \chi n_i} = \beta(1+r)\,\frac{1}{c_{j'}^2} + \xi_i, \qquad \xi_i\, b_i' = 0,$$

which for savers ($\xi_i = 0$), under logarithmic old utility (warm glow $\omega_B \log e$ included) and interior old choices, collapses to a savings rule:

$$b_i' = \beta\,(1+\gamma+\omega_B)\,(c_i - \chi n_i) + \mathbf{1}\{m = O\}\,\frac{\bar{\ell}\, h_i}{1+r},$$

with γ the old housing weight and ω_B the warm-glow weight: the household saves old age's claim on adult consumption—$\Lambda \equiv \beta(1+\gamma+\omega_B)$—plus her discounted future tax bill, so anticipating the lock-in tax is itself a savings motive. By the same conversion, a buyer pays an effective price $q^O = q + \bar{\ell}/(1+r)$ per unit of space—the market price plus her own discounted lock-in tax, pre-funded through savings rather than paid as a flow—while a renter pays q; write q^m for the mode's effective price. A unit of adult consumption commits $1+\Lambda$ units of resources, one spent now and Λ carried.

Then, with no binding constraints, with lifetime resources $w_i = y_i + b_i$,

$$c_i - \chi n_i = \frac{w_i}{1+\alpha+\vartheta+\Lambda}, \qquad n_i = \frac{\vartheta\, w_i}{(1+\alpha+\vartheta+\Lambda)(\chi+\kappa q^m)}, \qquad h_i = \frac{\alpha\, w_i}{(1+\alpha+\vartheta+\Lambda)\, q^m} + \kappa n_i,$$

Dividing the housing derivative of $\mathcal{L}$ by the consumption derivative, $\lambda_i = 1/(c_i - \chi n_i)$, prices every multiplier in goods. For a renter this gives

$$\frac{\alpha\,(c_i^R - \chi n_i^R)}{h_i^R - \kappa n_i^R} = q + \zeta_i^R, \qquad \zeta_i^R = \frac{\theta_i}{\lambda_i} \geq 0:$$

the left side is u_h/u_c, her marginal rate of substitution between space and consumption—the goods she would give up for one more unit of space, holding utility fixed—and it exceeds the rent exactly by the bite of the rental size limit, computable from her observed bundle. For a young owner the same two derivatives give

$$\frac{\alpha\,(c_i^O - \chi n_i^O)}{h_i^O - \kappa n_i^O} = q^O + \zeta_i^O, \qquad \zeta_i^O = \underbrace{\frac{\mu_i}{\lambda_i}(1-\phi)P}_{\zeta_i^{O,F},\text{ down payment}} + \underbrace{\frac{\theta_i}{\lambda_i}}_{\text{size limit}}:$$

her marginal value of space exceeds her effective price by the bite of whichever access constraints bind—the down payment (μ_i) or the owner size limit (θ_i). When the size limit is slack, the bundle identifies the financing bite alone (F for financing): $\zeta_i^{O,F}$ is the observed ratio minus q^O. For an old incumbent at an interior retention choice,

$$\frac{\gamma c_j^2}{h_j^2} = q - \bar{\ell}:$$

she keeps marginal space she values below what the market would pay, because selling it would realize the taxable gain. Old renters carry no such wedge: their marginal value is q.

The solved policies give the level of fertility; the first-order condition behind them shows where housing enters it. Dividing that condition by the marginal utility of consumption prices the marginal child in goods: in either tenure,

$$\underbrace{\frac{\vartheta\,(c_i^m - \chi n_i^m)}{n_i^m}}_{\text{willingness to pay for a marginal child}} = \underbrace{\chi + \kappa\,(q^m + \zeta_i^m)}_{\text{full price of a child}}.$$

A child's space requirement is priced at the household's own shadow value of space, so a binding access constraint acts as a tax of $\kappa\,\zeta_i^m$ per child. Children get more expensive exactly where family-sized space is hard to reach, with market prices unchanged.

3.5 Efficiency

I first ask whether the housing frictions described above can produce an inefficient allocation of family-sized space. To isolate this question, I consider a planner that faces the housing stock, size limits, goods feasibility, and the domain restrictions, but can use lump-sum transfers to support an allocation. The planner therefore does not face private borrowing constraints, which restrict purchases rather than the physical feasibility of occupying a home.

Proposition 1. *Hold the stationary cross-section, entry, tenure assignments, and savings fixed; the variation below resizes two owner-occupied houses and converts no tenures. The planner's surplus from moving one unit of space from an old incumbent at an interior retention choice to a young owner whose owner size limit is slack and whose financing constraint binds ($\zeta_i^{O,F} > 0$) is*

$$\underbrace{\left(q^O + \zeta_i^{O,F}\right)}_{\textit{buyer's marginal value}} - \underbrace{\left(q - \bar{\ell}\right)}_{\textit{incumbent's marginal value}} = \zeta_i^{O,F} + \frac{\bar{\ell}}{1+r} + \bar{\ell} > 0.$$

If this compensated local reassignment is feasible at zero real implementation cost and the displayed surplus is positive, the competitive allocation does not solve the planner's conditional allocation problem.

The proposition identifies a possible gain from moving space from an old owner to a constrained young buyer. The buyer values another unit at her effective price plus the financing gap, while the incumbent values the unit at the net-of-tax sale price. The market price cancels in the difference, and the tax payments are transfers rather than resource costs. This does not imply that every housing subsidy is welfare improving. Any real cost of relaxing the borrowing constraint must be deducted from the surplus, and a constraint that policy cannot relax creates no implementable gain.

Figure 1 illustrates this comparison. The buyer's marginal value of space lies above her effective price by $\zeta^{O,F}$, while the incumbent's value lies below the market price by $\bar{\ell}$. Compensating the incumbent at $q - \bar{\ell}$ and charging the buyer $q^O + \zeta^{O,F}$ leaves both households as well off as before and generates $\zeta^{O,F} + \bar{\ell}/(1+r) + \bar{\ell}$ in surplus.

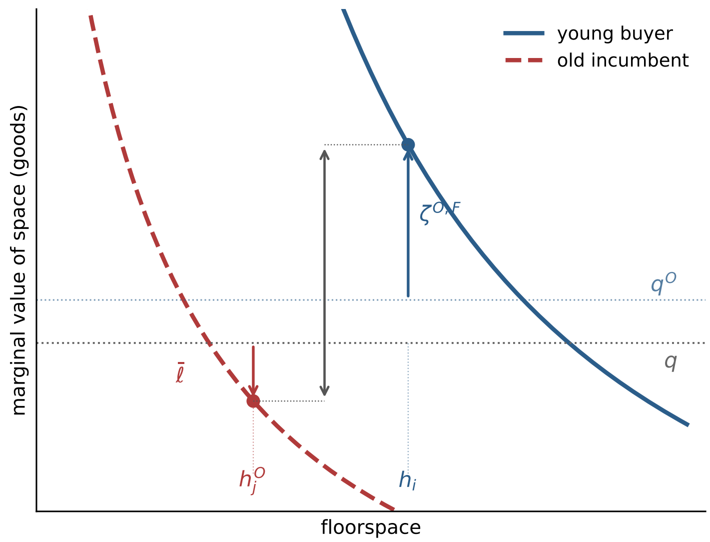


Figure 1: Misallocation. The collateral-capped buyer values marginal space at $q^O + \zeta^{O,F}$, above the market price; the locked-in incumbent values it at $q - \bar{\ell}$, below; the distance between the two points is the recoverable surplus.

3.6 Fertility and policy

Fix a tenure mode and a household whose cap binds, so that her housing is pinned at the effective family-housing cap defined in the household problem: the rental size limit for a renter, the shorter of the owner size limit and the collateral bound for a buyer. Write H for the level of that cap, $w = y + b$ for lifetime resources, and q^m for the mode's effective price. We relax the cap by a unit of space and study fertility. Concentrating the savings rule into the budget, fertility at the cap solves $\vartheta/n = (1+\Lambda)\chi/(w - q^m H - \chi n) + \alpha\kappa/(H - \kappa n)$; differentiating with respect to the cap level,

$$\frac{\mathrm{d}n}{\mathrm{d}H} = \frac{\dfrac{\alpha\kappa}{(H-\kappa n)^2} - \dfrac{(1+\Lambda)\,\chi q^m}{(w - q^m H - \chi n)^2}}{\dfrac{\vartheta}{n^2} + \dfrac{(1+\Lambda)\,\chi^2}{(w - q^m H - \chi n)^2} + \dfrac{\alpha\kappa^2}{(H-\kappa n)^2}},$$

which is positive for every binding cap if and only if $\chi \leq (1+\Lambda)\,\kappa q^m/\alpha$: relaxing access raises fertility whenever the child's consumption requirement does not exceed its space requirement priced at the lifetime cost of adult consumption. The derivative races two forces: extra space makes the child's space floor κ easier to meet, pushing fertility up, while paying q^m for the space tightens the budget around the child's goods floor χ, pushing fertility down. The savings margin weakens the second force: the housing bill is paid partly by saving less, so only a fraction $1/(1+\Lambda)$ of it falls on current spending, and the threshold below which access raises fertility scales up by $(1+\Lambda)$. Figure 2 draws the relation: fertility rises along the binding cap at rate $\mathrm{d}n/\mathrm{d}H$ and flattens at h^u, where the cap stops binding.

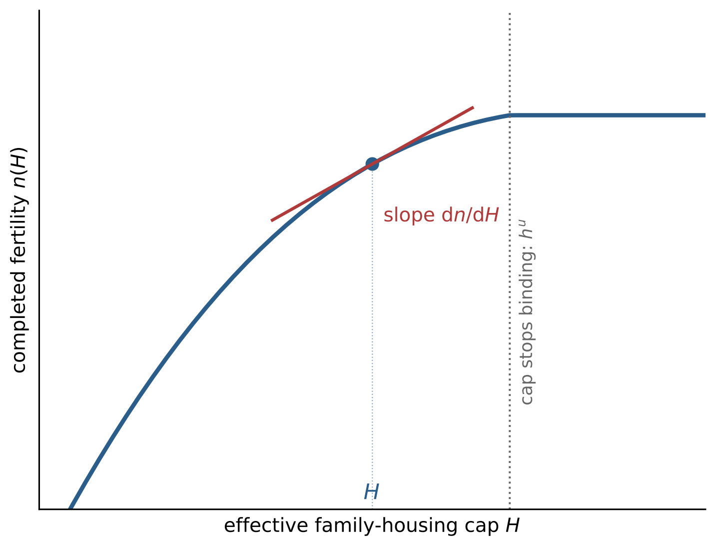


Figure 2: The fertility margin. Completed fertility against the effective family-housing cap, from the same solved case as Figure 1; the slope at the equilibrium cap is $\mathrm{d}n/\mathrm{d}H$, and the cap stops binding at h^u.

I next ask how a housing policy that relaxes these constraints affects family size. The aim is to improve the allocation of family-sized space, not to target fertility directly. Fertility changes only because the policy changes the cost of accommodating children. A down-payment grant raises b_i, looser credit raises ϕ, and either change relaxes the collateral bound $b_i\rho/((1-\phi)q)$. Making larger units available to rent raises $\bar{h}^R$. These policies leave the aggregate stock $\bar{H}$ unchanged and change who can occupy it. Let $\bar{n}_i^E$ denote the number of children in the outside option, normalized to zero in the baseline. At fixed prices, the effect of a small policy x on aggregate fertility is

$$\left.\frac{\mathrm{d}N}{\mathrm{d}x}\right|_q = \bar{M}\sum_m \int_{\mathcal{E}_m(x)} \left[\pi_i^E\,\Psi_i^m + (n_i^m - \bar{n}_i^E)\,\frac{\pi_i^E(1-\pi_i^E)}{\kappa_E}\,\Theta_i^m\right]\mathcal{P}_i^m(x)\,\mathrm{d}G_Y(i),$$

where $\mathcal{E}_m(x)$ is the set of households whose cap binds in mode m, $\mathcal{P}_i^m = \partial H_i^m/\partial x$ is the pass-through of the policy to the binding component of the cap, $\Psi_i^m = \mathrm{d}n/\mathrm{d}H$ is the fertility response, and $\Theta_i^m = \zeta_i^m/(c_i^m - \chi n_i^m)$ drives the entry term. A policy aimed at a slack constraint moves nothing: a down-payment grant does nothing for a household pinned by the rental size limit. At $\chi = (1+\Lambda)\kappa q^m/\alpha$, equal scaling at lifetime prices, Ψ_i^m is proportional to $\zeta_i^m/(c_i^m - \chi n_i^m)$, so the same gap that measures misallocation predicts whose fertility responds.

Policies can also change tenure. A change in $W_i^O - W_i^R$ moves some households across the rent–own boundary, and the gains tax affects this choice because a buyer anticipates the tax due upon a future sale.

4 Quantitative Lifecycle Model

4.1 Lifecycle and Income

The quantitative model asks whether access to housing affects family formation when households can save toward a purchase, move between renting and owning, and adjust their housing over the lifecycle. Households face income risk, make a family-size choice during the fertile years, and eventually leave bequests. Children raise the demand for space while they live at home, but older households may retain large homes after their children leave. The housing market has one user cost and one asset price. Tenure matters because it changes the available house sizes, financing constraints, and transaction costs.

Time is discrete. Households enter adult life at age $a = 1$, retire after age J_R, and die at the end of age J. Before retirement, earnings depend on age and idiosyncratic productivity $z \in \mathcal{Z}$. Income is

$$y(a,z) = \begin{cases} (1-\tau_{\text{pay}})we_a z, & a \le J_R, \\ p^{\text{ret}}, & a > J_R, \end{cases} \qquad z' \sim \Pi^z(\cdot|z), \tag{1}$$

The wage is common apart from age and productivity because the quantitative mechanism is housing access, tenure, and lifecycle wealth.

4.2 Preferences and Bequests

Children raise minimum consumption and housing needs while they live at home. Let n_s be the number of dependent children and n completed family size; the two differ after children mature and leave. Household needs are

$$\begin{aligned} \bar{c}(n,s) &= \bar{c}_0 + \bar{c}_n n_s, \\ \bar{h}(n,s) &= \bar{h}_0 + \bar{h}_{\text{jump}} \mathbf{1}\{n_s \ge 1\} + \bar{h}_n n_s. \end{aligned} \tag{2}$$

The first child can create a discrete housing-space jump; additional dependent children raise the required space linearly.

Preferences are CRRA over a Stone–Geary composite of consumption and usable housing space. Write h^R for rented space and h for the owned housing stock; only one can be positive. The owner service premium applies after household needs are removed:

$$u(c, h^R, h; n, s) = \frac{\left[(c - \bar{c}(n,s))^{\alpha_c} \mathfrak{h}(h^R, h; n, s)^{1-\alpha_c}\right]^{1-\sigma}}{1-\sigma} + \psi_{\text{child}} n_s, \tag{3}$$

where

$$\mathfrak{h}(h^R, h; n, s) = \begin{cases} h^R - \bar{h}(n,s), & h = 0, \\ \chi_O\left[h - \bar{h}(n,s)\right], & h > 0, \end{cases} \qquad \chi_O > 0. \tag{4}$$

Utility is $-\infty$ when residual consumption or usable space is nonpositive. When $\chi_O > 1$, ownership provides more utility from each unit of space left after household needs are met. It does not change the physical number of rooms required by children or the quantity of housing counted in market demand.

At death, households receive warm-glow bequest utility. The house passes to the estate at market value: there is no forced sale, no transaction cost, and no tax at death, so bequeathable wealth is

$W^B = b + Ph$, liquid wealth plus the gross value of any house held. Bequest utility is

$$\mathcal{B}(W^B, n) = \theta_0(1 + \theta_n n)\frac{(\theta_1 + \max\{W^B, 0\})^{1-\sigma}}{1-\sigma}. \tag{5}$$

The bequest term is not recentered at zero wealth. As a result, θ_n changes both the value of leaving a larger estate and the level of bequest utility across family sizes. Completed family size can therefore affect saving even after children have left the household.

4.3 Housing, Finance, and Taxes

There is one aggregate housing-services market. The user cost q and asset price P satisfy

$$q = (r + \delta + \tau^p)P. \tag{6}$$

Housing supply is upward sloping:

$$H^S(q) = \bar{H}\left(\frac{q}{\bar{q}}\right)^{\eta}. \tag{7}$$

Here $\bar{q}$ is the reference user cost, so housing supply equals $\bar{H}$ at $q = \bar{q}$. The fixed-stock compact model is the limiting case $\eta = 0$ around the benchmark stock. More generally, η determines whether an increase in housing demand appears mainly as a higher user cost or as additional housing.

Households either rent housing services or own one of K discrete housing sizes. The owned housing stock h belongs to

$$\mathcal{H} = \{0, H_1, \ldots, H_K\}, \tag{8}$$

where $h = 0$ denotes a renter and $h = H_k$ an owner of a house of size H_k, ordered so that $H_1 < \cdots < H_K$. Thus tenure is implicit in the housing position rather than carried by a separate indicator. Renters choose h^R continuously up to $\bar{h}^R$; owners choose from the discrete ladder. The ladder is the quantitative version of the family-housing segmentation in Section 3.

Renters choose $h^R \in [0, \bar{h}^R]$ and cannot borrow. Owners choose from the discrete ladder and can borrow against the house. They pay maintenance and property taxes through $(\delta + \tau^p)PH_k$.

At origination, a buyer can finance at most a share ϕ of house value, so the required down payment is

$$(1 - \phi)PH_k. \tag{9}$$

Equivalently, adjusted liquid wealth immediately after purchase must satisfy $\tilde{b} \geq -\phi PH_k$. End-of-period saving for every owner, whether a buyer or an incumbent, faces the same debt floor:

$$b' \geq -\phi PH_k. \tag{10}$$

The down payment governs entry into ownership, while the debt floor governs resources after purchase. The distinction matters because a household can be able to service an owned home and still lack the liquid wealth needed to buy it.

Housing adjustment is costly. A sale of an owned house $h > 0$ incurs the proportional transaction cost ψ^s, so net sale proceeds are

$$\Omega(h) = (1 - \psi^s)Ph. \tag{11}$$

Transaction costs are the only disposal friction in the computed benchmark: an owner who sells pays ψ^s per unit of value and faces no tax on accrued gains.

The quantitative exercises do not yet include the capital-gains treatment of Section 3. Adding it requires tracking a tax basis B^p for each owner. A sale would then trigger $T^g(PH_k, B^p) = \tau^g \max\{PH_k - B^p - \bar{g}, 0\}$ with statutory exclusion $\bar{g}$, net proceeds would fall to $\Omega(H_k) - T^g(PH_k, B^p)$, and stepped-up basis at death would forgive the accrued gain. The benchmark instead captures selling costs through ψ^s and studies old-owner retention with a direct policy experiment.

4.4 Fertility and Child Aging

Fertility is chosen during the fertile window

$$a \in \{A_f^{\text{start}}, \ldots, A_f^{\text{end}}\}.$$

A childless household chooses

$$n' \in \mathcal{N} = \{0, 1, \ldots, \bar{n}\}. \tag{12}$$

The option $n' = 0$ means no birth in the current period. A positive choice fixes completed fertility forever and moves its children into the first dependent stage. A household that reaches the end of the fertile window without a positive choice remains childless.

In the computation, $n \in \{0, 1, 2\}$ is a family-size index rather than a literal count of children. The category $n = 1$ represents families with one or two children, while $n = 2$ represents families with three or more. The maintained scaling assigns two children to each unit of the index, so the completed-fertility measure matched to the CPS is reported as $2\mathbb{E}[n]$. Preferences, child needs, and bequests are evaluated using the index n; the factor of two is applied only when reporting completed fertility.

Let $s = 0$ denote no dependent children and let $s = 1, \ldots, S_c$ denote dependent-child stages. After a positive fertility choice, the household enters $s = 1$. Children age stochastically through the stages according to Π^c. While children are at home, $n_s = n$; after maturity, $n_s = 0$. Completed fertility still matters through bequest utility.

4.5 Household Problem

Within the period, the household first faces fertility if eligible, then chooses its owned housing stock h' or rental services $h^{R\prime}$, and then chooses consumption and saving. A renter sets $h' = 0$; an owner sets $h^{R\prime} = 0$. Let the continuation value after income risk and child aging be

$$\bar{V}(b', z, h', a+1, n, s) = \sum_{z'} \sum_{s'} \Pi^z(z'|z) \Pi^c(s'|s, n) V(b', z', h', a+1, n, s'). \tag{13}$$

The terminal value is

$$V(b, z, h, J+1, n, s) = \mathcal{B}(b + Ph, n).$$

Conditional on renting, the branch value is

$$\begin{aligned} V^R(\tilde{b}, z, a, n, s) = \max_{c, h^{R\prime}, b'} \quad & u(c, h^{R\prime}, 0; n, s) + \beta \bar{V}(b', z, 0, a+1, n, s) \\ \text{s.t.} \quad & c + q h^{R\prime} + b' = (1+r)\tilde{b} + y(a, z), \\ & 0 \le h^{R\prime} \le \bar{h}^R, \qquad b' \ge 0. \end{aligned} \tag{14}$$

Conditional on choosing an owned house $h' > 0$ after any adjustment, the branch value is

$$\begin{aligned} V^O(\tilde{b}, z, h', a, n, s) = \max_{c,b'} \quad & u(c, 0, h'; n, s) + \beta \bar{V}(b', z, h', a+1, n, s) \\ \text{s.t.} \quad & c + (\delta + \tau^p)Ph' + b' = (1+r)\tilde{b} + y(a, z), \\ & b' \geq -\phi Ph'. \end{aligned} \tag{15}$$

The debt floor is the same for a buyer and an incumbent. Purchase feasibility is screened separately through the down-payment condition before this branch is evaluated. Renting offers a continuous choice of space but no borrowing; ownership offers the larger discrete menu and borrowing against the house, at the cost of the initial down payment and future adjustment costs.

The liquid wealth available after changing the owned housing stock from h to h' is

$$\tilde{b}(b, h, h') = b + \mathbf{1}\{h > 0,\ h' \neq h\}\Omega(h) - \mathbf{1}\{h' > 0,\ h' \neq h\}Ph'. \tag{16}$$

A seller collects net proceeds, a buyer pays the full purchase price, and a household that keeps its position pays nothing.

The housing-position value for a feasible $h' \in \mathcal{H}$ is

$$\widetilde{V}^H(h'|b, z, h, a, n, s) = \begin{cases} V^R(\tilde{b}(b, h, 0), z, a, n, s), & h' = 0, \\ V^O(\tilde{b}(b, h, h'), z, h', a, n, s), & h' \in \mathcal{H} \setminus \{0\}. \end{cases} \tag{17}$$

An infeasible branch is assigned value $-\infty$. The rent-or-own and owner-size choices have Type-I extreme-value shocks with scale κ_T:

$$\pi^H(h'|b, z, h, a, n, s) = \frac{\exp\{\widetilde{V}^H(h'|b, z, h, a, n, s)/\kappa_T\}}{\sum_{\hat{h} \in \mathcal{H}} \exp\{\widetilde{V}^H(\hat{h}|b, z, h, a, n, s)/\kappa_T\}}, \tag{18}$$

and the inclusive housing value is

$$V^H(b, z, h, a, n, s) = \kappa_T \log \sum_{\hat{h} \in \mathcal{H}} \exp\{\widetilde{V}^H(\hat{h}|b, z, h, a, n, s)/\kappa_T\}. \tag{19}$$

At the estimated value $\kappa_T = 0$, these expressions mean their deterministic limit: the branch with the highest value is chosen with probability one, and a tie is resolved by the fixed ordering of feasible branches.

For a childless household in the fertile window, the value of fertility option n' is

$$\widetilde{V}^F(n'|b, z, h, a, 0, 0) = V^H(b, z, h, a, n', s(n')), \qquad s(n') = \mathbf{1}\{n' > 0\}. \tag{20}$$

With Type-I extreme-value shocks of scale κ_F,

$$\pi^F(n'|b, z, h, a, 0, 0) = \frac{\exp\{\widetilde{V}^F(n'|b, z, h, a, 0, 0)/\kappa_F\}}{\sum_{\hat{n} \in \mathcal{N}} \exp\{\widetilde{V}^F(\hat{n}|b, z, h, a, 0, 0)/\kappa_F\}}. \tag{21}$$

The value before the fertility draw is the corresponding inclusive value:

$$V(b, z, h, a, 0, 0) = \kappa_F \log \sum_{\hat{n} \in \mathcal{N}} \exp\{\widetilde{V}^F(\hat{n}|b, z, h, a, 0, 0)/\kappa_F\}. \tag{22}$$

Outside the fertile window, or after a positive fertility choice, this stage is degenerate and $V(b, z, h, a, n, s) = V^H(b, z, h, a, n, s)$. The family-size decision therefore incorporates the housing position chosen in the same period. A birth can change current space and tenure immediately, not only the household's continuation value.

4.6 Stationary Equilibrium

Let g denote the beginning-of-period stationary composition of adult households over liquid wealth, productivity, inherited owned housing, age, family size, and child age, (b, z, h, a, n, s), normalized to integrate to one. Total adult scale is S, so the corresponding level measure is $G = Sg$:

$$\int dg(b, z, h, a, n, s) = 1, \qquad \int dG(b, z, h, a, n, s) = S. \tag{23}$$

The distinction between g and G separates composition from scale: household policies determine the composition of the population, while entry and replacement determine how many adults the modeled market contains.

Definition 1 (Stationary equilibrium)**.** A stationary equilibrium is a user cost and asset price (q, P), value functions and policies, housing and fertility choice probabilities, stationary composition g, adult scale S, level measure $G = Sg$, and aggregate housing quantities such that:

1. households solve the branch problems (14)–(15), housing choice is given by (18), fertility choice is given by (21), and terminal utility is (5);
2. current choices, saving, income risk, child aging, adult aging, death, and the injection of new entrants induce a normalized transition operator $\mathcal{T}(g; q, S)$. Stationarity requires

$$g = \mathcal{T}(g; q, S); \tag{24}$$

3. let $\Pi^H(h' \mid b, z, h, a, n, s)$ and $\widehat{h}^R(b, z, h, a, n, s)$ denote the housing-position probability and expected rental space conditional on renting, respectively, after integrating over the current family-size choice. Per-unit physical housing demand is

$$\begin{aligned} \widehat{H}^D(g, q) = \int \Bigg[& \Pi^H(0 \mid b, z, h, a, n, s)\widehat{h}^R(b, z, h, a, n, s) \\ & + \sum_{k=1}^{K} \Pi^H(H_k \mid b, z, h, a, n, s) H_k \Bigg] \mathrm{d}g(b, z, h, a, n, s). \end{aligned} \tag{25}$$

4. let $E_0(g, q)$ be the inflow of young adults required to reproduce one unit of the stationary composition, $B_0(g, q)$ the mature locally born flow per unit of adult scale, M the outside-origin entrant pool, and $q^E(q)$ the probability that a potential entrant enters the modeled market. Entry balances adult replacement when

$$SE_0(g, q) = q^E(q) \left[M + SB_0(g, q) \right]. \tag{26}$$

   Provided $E_0(g, q) > q^E(q)B_0(g, q)$, this condition gives the finite scale

$$S = \frac{q^E(q)M}{E_0(g, q) - q^E(q)B_0(g, q)}; \tag{27}$$

5. the housing market clears in levels and the user-cost relation holds:

$$S\widehat{H}^D(g, q) = H^S(q), \qquad q = (r + \delta + \tau^p)P; \tag{28}$$

6. taxes, pensions, and any rebates satisfy the government accounting rule specified for the calibration or counterfactual.

Equilibrium links family formation to housing through two margins. Children change housing demand within a household, and, when locally born children feed the future entrant pool, they also change the scale of the market. Housing supply determines how much of either margin is absorbed by quantities rather than by the user cost.

The benchmark normalizes $S = 1$. In the counterfactuals with entry, population and the housing price adjust jointly: a policy can attract households into the market, and the additional housing demand feeds back into the user cost.

4.7 Computation

One model period is four years. Period $j = 0, \dots, 16$ begins at age $18 + 4j$; the last interval covers ages 82–85. Retirement begins at 66, and the last period in which a birth can occur begins at 42. Annual parameters are converted before solution:

$$1 + r = (1 + r_{\text{ann}})^4, \quad \beta = \beta_{\text{ann}}^4, \quad \delta = 1 - (1 - \delta_{\text{ann}})^4, \quad \tau^p = 4\tau^p_{\text{ann}}.$$

Income and consumption-flow quantities are expressed in four-year units; housing remains in rooms.

The liquid-wealth grid has $N_b = 120$ nodes on $[-12, 30]$, with a dense core on $[-5, 7]$ and a thinner upper tail beyond 15. Income takes values $z \in \{0.6, 0.8, 1.0, 1.2, 1.4\}$ with invariant weights $\pi_z = (0.1, 0.2, 0.4, 0.2, 0.1)$ and transition matrix $\Pi^z = 0.85I + 0.15\mathbf{1}\pi_z'$. One dependent-child stage has an expected duration of 18 years, giving a four-year maturity probability of $2/9$. Entrants have mass $1/J$ each period. The balanced pension equals payroll-tax revenue per retiree in the stationary age distribution.

I solve the household problem by backward induction and obtain the stationary distribution by forward iteration. The user cost adjusts until aggregate housing demand equals supply, and I exclude any solution with a market residual above 10^{-4}.[7]

Grid resolution matters because tenure and family-size choices are nearly deterministic at the estimated parameters. Small changes in the wealth grid can move households across the purchase threshold. An earlier calibration using $N_b = 60$ fit 18 percent worse when evaluated at $N_b = 120$, showing that part of its apparent fit was numerical. I therefore use $N_b = 120$ for the estimation and all policy exercises. A separate evaluation at $N_b = 240$ is reported below only as a sensitivity check.

5 Calibration

5.1 Parameters Set Outside the Estimation

One model period corresponds to four years. I take the financial parameters and income process from outside the model, choose the available house sizes to match those observed in the ACS, and measure the initial wealth distribution of entrants directly in the PSID.

[7] Saving within each branch is solved by golden-section search; Howard iteration accelerates value-function evaluation, and a damped fixed point followed by scalar bracketing solves for the housing user cost.

Table 1: Parameters set outside the estimation

Parameter	Value	Source or rule
Interest rate r	4% per year	standard
Depreciation δ	2% per year	standard
Property tax τ^p	1% per year	U.S. average effective rate
Loan-to-value ratio ϕ	0.80	20% down payment
Sale cost ψ^s	0.06	transaction-cost evidence
Risk aversion σ	2	standard
Bequest-utility shift θ_1	0.01	near-linear warm glow
Payroll tax τ_{pay}	0.179	pension balances in the stationary distribution
Income profile e_a	0.65 at 18, rises to 1.0 at 34, declines after 58	normalized lifecycle profile; five-state persistent income process
Owner house sizes $\{H_k\}$	$\{2, 4, 6, 8, 10\}$ rooms	room-count support, ACS
Largest rental unit $\bar{h}^R$	6.0 rooms	90th-percentile rental size rule of Greaney et al. (2025), measured in the matched ACS sample; 6.5 as robustness
Housing supply $(\bar{H}, \eta)$	$(4, 1)$	unit-elastic benchmark
Entrant wealth distribution	PSID young-renter wealth-to-income distribution	measured among young renters in the PSID

I set the largest available rental unit to six rooms, corresponding to the upper tail of the rental-size distribution in the matched ACS sample. The fit changes little when I raise this value to 6.5 rooms and deteriorates outside that range. I choose the owner house sizes from the observed support of rooms in the ACS. The data therefore determine which housing products are available, while the estimation determines how households value them and sort across them.

5.2 Estimated parameters

The remaining parameters govern saving, housing demand, tenure choice, family size, and bequests. I estimate their 13 values using 14 moments on family size, housing consumption, tenure, and lifecycle wealth. The estimates minimize a weighted distance between the model and the data using $N_b = 120$ points for liquid wealth. Fourteen moments provide one overidentifying restriction by count. In practice, several parameters have similar effects on the moments, so not all of them are precisely identified.[8]

[8] I first search globally using differential evolution and then refine the best candidates with local minimization.

Table 2: Estimated parameters in the July 9 calibration

Parameter	Interpretation	Estimate
β_{annual}	Annual discount factor	0.940
α_c	Consumption weight	0.620
$\bar{c}_0$	Baseline consumption requirement	1.279
$\bar{c}_n$	Consumption cost per dependent child	0.116
$\bar{h}_0$	Baseline housing requirement	1.000
$\bar{h}_{\text{jump}}$	Discrete housing-need jump at the first child	2.120
$\bar{h}_n$	Housing requirement per dependent child	1.382
ψ_{child}	Utility from children	0.349
κ_F	Dispersion in family-size choices	1.019
κ_T	Dispersion in tenure choices	0.000
χ_O	Utility premium from owner-occupied housing	1.150
θ_0	Bequest strength	0.0015
θ_n	Effect of family size on the bequest motive	0.972

Lifecycle wealth and homeownership profiles identify the saving parameters. The discount factor β primarily changes asset accumulation, while the baseline consumption requirement $\bar{c}_0$ reduces the resources available for saving, housing, and children. I use liquid wealth among young renters, old-age nonhousing wealth, and homeownership at young and old ages to separate these effects. The separation remains weak because both parameters influence tenure through accumulated wealth.

I identify housing needs from how housing consumption changes with family composition. Mean rooms among childless renters identifies the baseline requirement $\bar{h}_0$; the change in rooms when the first child arrives identifies $\bar{h}_{\text{jump}}$; and the difference in rooms between larger and smaller families identifies the additional requirement $\bar{h}_n$. These moments also respond to income, wealth, and tenure, so the three estimates are strongly correlated.

The consumption weight α_c changes housing demand in both tenure arrangements, while χ_O changes the value of owner housing relative to rental housing. I use the owner–renter room difference, the share of owners in large homes, and homeownership by age and family status to distinguish the two. The distinction is imperfect because wealth and borrowing constraints also affect whether a household owns.

Completed fertility and childlessness identify the consumption cost of children $\bar{c}_n$, the utility value of children ψ_{child}, and the dispersion of family-size choices κ_F. The three parameters have different economic roles: $\bar{c}_n$ reduces the resources left for other goods, ψ_{child} raises the direct value of children, and κ_F determines how dispersed choices are across otherwise similar households. Nevertheless, these effects can substitute for one another locally. Homeownership by age and family status identifies κ_T. Its estimate of zero means that additional random variation in tenure choice does not improve the fit; it does not mean that the parameter is precisely estimated.

Finally, I use old-age wealth and homeownership to identify the overall strength of the bequest motive θ_0, and the parent–childless difference in old-age wealth to identify how that motive varies with family size through θ_n. The second effect is weakly identified because the estimated overall bequest motive is close to zero. I therefore treat the estimates as a current calibration, not as evidence that every preference parameter is precisely estimated.[9]

[9] Several estimates lie at or near the limits of the July search region.

5.3 Target moments and fit

The calibration targets fourteen moments: completed fertility and childlessness from the CPS (the June 2024 supplement cells for women 40–44), tenure and housing-size moments by age and family status from a matched ACS metropolitan sample, and wealth moments from the PSID. Table 3 compares the data with the model at the July 9 calibration. The weighted objective is 6.31.

Table 3: July 9 target moments and model fit ($N_b = 120$; loss 6.31)

Moment	Data	Model
Old-age ownership rate (65–75)	0.764	0.893
Ownership rate (25–34)	0.341	0.226
Owner–renter mean rooms (no children at home, 30–55)	2.419	2.151
Family ownership gap	0.168	0.298
Young renter liquid wealth / income (25–35)	0.179	0.353
Ownership rate (30–55)	0.575	0.531
Renter mean rooms (no children at home, 30–55)	3.805	3.655
Childlessness rate	0.188	0.260
Rooms gap, 3+ vs. 1–2 children (30–55)	0.368	0.269
Old parent–childless nonhousing wealth gap	1.007	0.842
Housing increment at first birth (rooms)	0.664	0.618
Completed fertility	1.918	1.949
Old nonhousing wealth / income, median (65–75)	2.230	2.328
Owner share with rooms ≥ 6 (no children at home)	0.596	0.593

The model is closest to the data on average completed fertility, the increase in rooms when the first child arrives, old-age nonhousing wealth, and the share of childless owners in large homes. It performs less well on the distribution of family size: childlessness is 0.26 rather than 0.19, even though average completed fertility is nearly exact. The family ownership gap is also too large, 0.30 rather than 0.17.

The lifecycle pattern of ownership reveals a second problem. Homeownership at ages 25–34 is too low, 0.23 rather than 0.34, while liquid wealth among young renters is too high, 0.35 rather than 0.18. Taken together, these two misses show that low early ownership is not simply caused by too little saving in the model. At the other end of the lifecycle, the model keeps 89 percent of households aged 65–75 in owned housing, compared with 76 percent in the data. Without health shocks, mortality risk, or another reason to downsize, leaving ownership is too rare.

The reported fit remains provisional for two reasons. First, the July calibration dates some purchases one period late. Correcting their timing changes several tenure and housing moments, including the housing response when the first child arrives. Second, holding the parameters fixed and raising the number of wealth points from 120 to 240 increases the loss from 6.31 to 6.84, mainly through the family ownership gap and old-age homeownership. The lifecycle and decision-rule figures below use the preliminary solution computed after correcting purchase timing. The remaining quantitative tables and policy exercises retain the July 9 values and will be regenerated after re-estimation.

6 Lifecycle Housing Patterns

Figure 3 reports lifecycle profiles from the preliminary solution computed after correcting the timing of current housing choices. Homeownership and housing consumption rise together through the years in which households form families. Between ages 30 and 42, homeownership rises from 23 to 71 percent and mean housing consumption rises from 4.3 to 5.7 rooms. Expected births are concentrated over the same part of the lifecycle and peak shortly before the end of the fertile window.

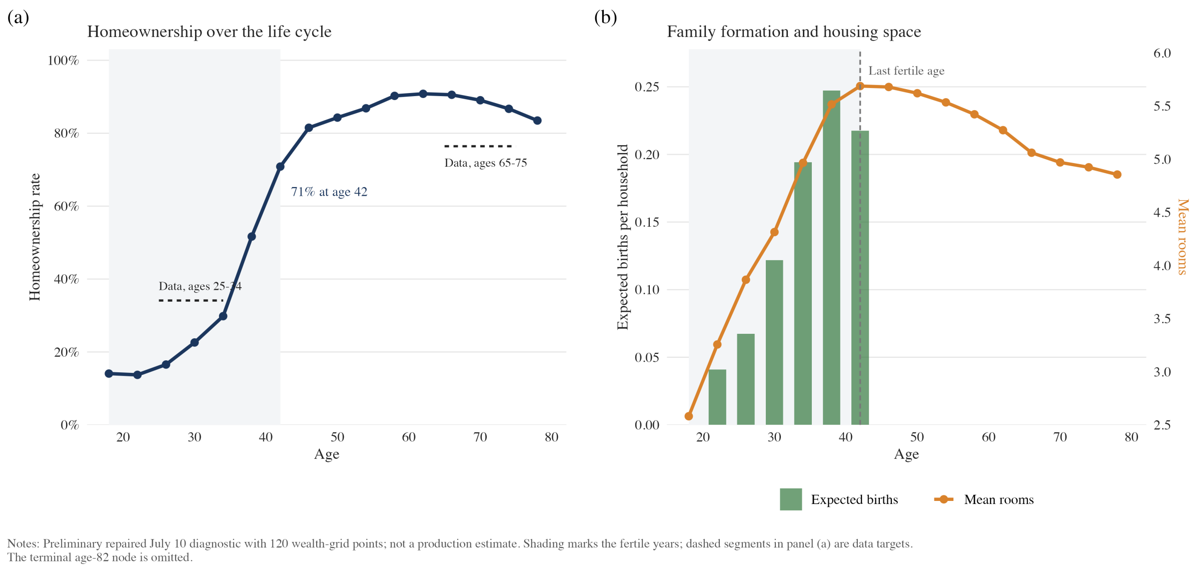
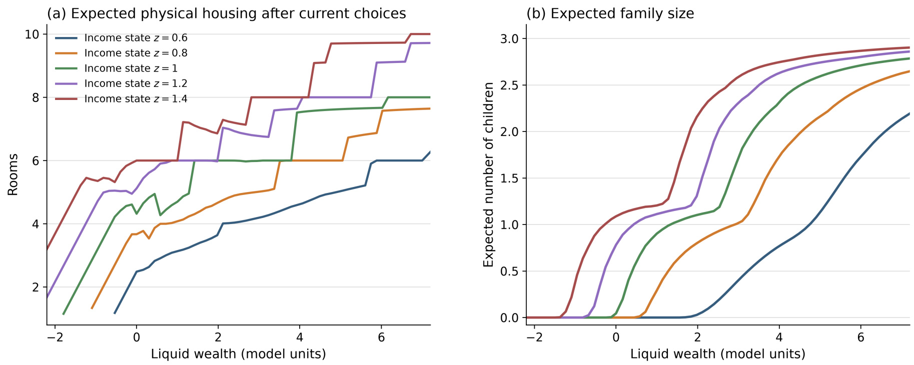


Figure 3: Lifecycle ownership, family formation, and housing consumption in the preliminary repaired solution. Panel (a) plots the homeownership rate and the young and old data targets. Panel (b) plots expected births per household and mean physical housing by age.

This co-movement does not by itself establish that housing causes births. It shows that the housing and family-size decisions are concentrated in the same lifecycle window, when wealth accumulation lets households move into ownership and up the housing ladder. The remaining fit problem is also clear. The rise in ownership is too steep: the model has too few young owners and retains too many older owners, while births remain too concentrated near the fertility deadline. I therefore use these profiles to describe the mechanism rather than as a final validation of the calibration.

The decision rules underlying these aggregates are shown in Figure 4. The figure considers households that enter age 42 as childless renters. Income shifts both rental demand and the owned house selected. At a given level of liquid wealth, higher-income households generally choose more rooms and a larger expected family size. Within an income state, additional wealth expands housing consumption and moves households through the discrete owner ladder.



Notes: Expected current-choice outcomes for households that enter age 42 as childless renters. Panel (a) averages branch-specific rooms over the family-size choice probabilities. Preliminary repaired July 10 diagnostic; not a production estimate.

Figure 4: Housing and family-size policies at the end of the fertile window. Panel (a) reports expected physical rooms after integrating over the current family-size choice and the housing choice that follows it. Panel (b) reports expected family size. Each line corresponds to a current income state.

The full housing schedules reveal heterogeneity that is hidden by a binary rent–own indicator. Several income states cross into ownership at the same wealth-grid point because they face a common down-payment rule, but they choose different rental quantities and different rungs of the owner ladder. Physical rooms can also fall locally when a household enters ownership. Such a move can still be optimal because ownership changes the utility from usable space, the household balance sheet, and continuation values; the number of rooms alone is not the household's payoff from owning.

7 Quantitative Evidence on the Mechanism

When a rental-size or financing constraint binds, a household would pay more for another room than the market user cost. I denote this excess willingness to pay by ζ. It answers a simple economic question: how much would the household gain from access to one additional room?

Among fertile-age renters choosing the largest available rental unit, the average value of ζ is 0.248 goods units, or 1.28 times the market user cost. These households therefore value another room at about 2.3 times its market price. To locate the households whose family-size choices are most sensitive to additional housing, I weight them by the response $\partial n/\partial H$ derived in Section 3. Seventy-three percent of this sensitivity is concentrated among households with $\zeta > 0$.

The model is motivated by the allocation of large homes across generations. The next table compares who owns homes with at least six rooms in the current calibration and in the ACS.

Table 4: Who holds the family-sized owner stock (rooms ≥ 6)

Holder	Data (ACS)	Model
Age 18–29	3.4%	0.4%
Age 30–45	24.8%	12.7%
Age 46–64	43.3%	50.5%
Age 65+	28.5%	36.5%
Held by households 46+	71.8%	87.0%
Held by households with no children at home	54.9%	72.5%

The concentration is present in the data, not only in the model. In the ACS, 72 percent of family-sized owner housing is held by households past the fertile years and 55 percent by households with no children at home. The model has the same ordering but overstates the concentration among older households. The data establish that older households hold a large share of family-sized homes; whether this allocation is inefficient depends on relative marginal valuations, which the model supplies.

8 Housing Policy

I next conduct a preliminary study of housing policies. The idea is to ask whether policies that move family-sized space toward households that value it more can reduce the inefficiency identified in the analytical model, and whether this reallocation changes family formation. I consider a holding tax on large owner-occupied homes, a birth-linked down-payment grant restricted to family-sized homes, greater access to large rental units, and a package that combines the tax and the grant. I also raise the selling cost faced by older owners to study the role of retention directly. These are exploratory exercises based on the July calibration, not policy estimates. Housing prices adjust to clear the market in each case. The calibration is useful for comparing the mechanisms, but the magnitudes are not final.

8.1 Policy Effects with Fixed Population

I first hold the adult population fixed so that the results reflect changes in the decisions of households already in the market.

A two percent holding tax on large owner-occupied homes works mainly through capitalization. The asset price of these homes falls by 12 percent, lowering the down payment required to buy them. Completed fertility, however, rises by only +0.004 to +0.009. The response is small because the households affected by the tax already hold wealth above the purchase threshold. A lower price changes which house they buy more than whether they have children.

Expanding the largest rental unit from six to 6.5 rooms also has essentially no effect on completed fertility. The policy gives constrained renters access to slightly more space, but it also makes renting more attractive relative to owning. In the current calibration, these effects offset one another.

The birth-linked down-payment grant has a larger effect. A grant paid at a birth and restricted to family-sized homes raises completed fertility by +0.01, +0.13, +0.26, and +0.49 at grant sizes 0.1, 0.4, 0.58, and 1.0, measured in units of mean annual income. The grant relaxes the financing constraint at the same time that the household is deciding whether to have children. This design also creates a waiting incentive: some households delay a purchase until a birth to qualify for the subsidy. Young ownership falls from 0.23 to 0.09 at the smallest grant, although the delay becomes smaller as the grant increases.

I next impose a selling cost equal to 20 percent of house value on older owners. This experiment strengthens the incentive to retain a home, but it changes the housing price by less than one percent and completed fertility by less than one hundredth. The small response reflects an important omission from the model. Without health shocks, mortality risk, or another reason to downsize, most old owners already keep their homes, so an additional selling cost changes little. The result therefore does not show that old-owner retention is unimportant in the data.

Finally, I combine the holding tax and the grant. The tax reduces the value of keeping a large home and lowers its price, while the grant helps a young household finance the purchase. In the July calibration, the two percent tax funds the grant, lowers the price of large homes by 11 percent, and raises completed fertility by +0.15.

8.2 Market Entry and Population

A housing policy can affect both family size among current residents and the number of households living in the market. The preceding exercises isolate the first response by fixing adult population at $S = 1$. I now allow entry to respond. A policy that makes the market more attractive draws in households from outside; if it changes births, it also changes the future flow of locally born entrants. Equation (26) determines population, and equation (28) determines the new housing price. I hold the attractiveness of the outside option and the pool of potential entrants fixed across policies. The calculation assumes that locally born children become future potential entrants one-for-one.

Table 5 reports the same experiments with entry allowed to respond. The second fertility column gives the change in completed fertility per household, while S gives adult population relative to the benchmark. All cases use $N_b = 120$.[10]

Table 5: Policy exercises with endogenous entry ($N_b = 120$)

Policy	Δ completed fertility, fixed	Δ completed fertility, entry	Population S	Δ total births
Two percent holding tax	+0.004	+0.002	1.003	+0.4%
Grant of 0.4 for homes with at least six rooms	+0.119	+0.083	1.038	+8.2%
Package: holding tax and grant	+0.162	+0.086	1.040	+8.6%
Maximum rental size raised to 6.5 rooms	−0.006	−0.009	1.004	≈ 0.0%
Twenty percent selling cost for older owners	+0.006	+0.000	1.007	+0.7%

Allowing entry attenuates the effect of the grant on families already in the market. The grant attracts more households, population expands, and the resulting increase in the housing price offsets part of the fixed-population fertility response. Total births nevertheless rise in the grant and package exercises. The holding tax, the increase in maximum rental size, and the higher selling cost for older owners remain close to neutral. The one-market model cannot determine whether the increase in local births is creation or relocation from elsewhere.

[10] Because population and the housing price are solved jointly, the grant and package rows retain a few percent of price-solving error.

9 Measurement and Re-estimation

In the July calculation, new purchases enter some aggregate outcomes one period late. This is inconsistent with the household problem, in which a buyer receives the new house immediately. I have corrected the measurement so that a current purchase contributes to current housing demand and to the housing response when the first child arrives. Because the correction changes several moments used in estimation, the model must be re-estimated.

The re-estimation uses the income profile, wealth grid, housing sizes, and target definitions described above, with 120 wealth points throughout. Three features of the economic specification do not change: the owner premium applies to usable space $H_k - \bar{h}(n, s)$, bequest utility varies with family size, and all owners face the same debt limit.

10 Limitations

The main omission concerns housing adjustment in old age. The model excludes health shocks, mortality risk, and other reasons to downsize, and transaction costs are the only reason an older owner is reluctant to sell. This pushes old-age homeownership upward and helps explain why the experiment that raises selling costs has little effect. The quantitative model also omits the capital-gains and basis mechanism studied in Section 3.

The timing of family formation is also simplified. There is no age-dependent fecundity, so postponing a birth is costless until the last fertile period. This feature contributes to the bunching of purchases and births near the end of the fertile years.

Finally, the quantitative exercises use unit-elastic housing supply, whereas the analytical reallocation result holds the stock fixed. The entry results also depend on the benchmark outside option and on the assumption that local births become future entrants. The one-market model therefore separates family size from local population growth, but it cannot identify a national migration or birth response.

11 Conclusion

Children use housing space, and access to that space differs sharply by tenure. A rental-size limit raises the shadow price of room for a constrained household. Ownership relaxes the size limit but introduces a down payment. Retention frictions can keep large homes with older households whose current use of space is low. The analytical model shows that these constraints raise the effective cost of accommodating children and can leave family-sized space with households that value it less.

The July calibration suggests that this mechanism is active. Many households deciding whether to have children value additional space above its user cost, purchases bunch late in the fertile years, and older households hold most family-sized owner housing in the data.

The policy exercises sharpen the economic distinction among interventions. A tax on large homes is capitalized into prices and down payments, but has little effect on family size because the affected buyers already hold enough wealth to purchase. Expanding the rental-size limit and raising old-owner selling costs also have small effects in the current calibration. A birth-linked grant has a larger effect because it relaxes the financing constraint when the household is making its family-size choice, although it also induces some households to delay a purchase and its effect is attenuated when entry raises housing demand and prices.

These magnitudes are not final because correcting purchase timing requires re-estimation, and the weak old-owner response reflects the model's limited downsizing margin. The current comparison nevertheless points to a simple lesson: housing policy affects family formation when it changes access to family-sized space for constrained households at the time they are deciding whether to have children. Policies that mainly change asset values or target households that are unlikely to adjust have much smaller effects.

A Additional Analytical Results

This appendix records three extensions to the short analytical note. They are not needed for the main argument in Section 3.

First, at equal scaling, $\chi = (1+\Lambda)\kappa q^m/\alpha$, the fixed-mode response to a marginal relaxation of a binding housing cap can be written as

$$\Psi_i^m = \frac{\kappa}{\alpha}\frac{\zeta_i^m}{c_i^m - \chi n_i^m}\,\omega_i^m, \qquad \omega_i^m > 0.$$

The normalized access gap therefore governs both the response of family size among incumbent households and the response of entry into the modeled market.

Second, for a collateral-capped owner, $H_i^O = b_i\rho/[(1-\phi)q]$, with $\rho = r + \delta + \tau^p$. Holding the user cost and the old-owner gains wedge fixed,

$$\left.\frac{\partial H_i^O}{\partial \tau^p}\right|_{q,\bar{\ell}} = \frac{b_i}{(1-\phi)q} > 0. \tag{29}$$

A property-tax increase can therefore relax the down-payment constraint by lowering the capitalized asset price, even when the housing user cost is held fixed.

Finally, a policy can move households across the rent–own boundary as well as relax a constraint within a tenure. The resulting change in births depends on both the mass of households near that boundary and the difference in family size between their owner and renter choices. Its sign is not determined by the direction of tenure switching alone.

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